Everything previously required a positive series. What about when it changes sign?

Corollary to absolute conv.

**Alternating Series Test:** If $\underbrace{a_n \geq a_{n+1}}_{\text{Series is decreasing}}$ (1) for all $n$ and $\lim_{n\to\infty} a_n = 0$ (2), then the alternating series

$$\sum_{n=1}^{\infty} (-1)^n a_n$$

converges.

If (1) and (2) are met, then $\sum_{n=1}^{\infty} (-1)^n a_n$ converges

**Absolute Convergence Test:** If $\sum_{i=1}^{\infty} |a_i|$ converges, then $\sum_{i=1}^{\infty} a_i$ also converges.

/Absolute converges theorem

↑ If you ignore changing signs & it converges, then w/ changing signs it also converges

will only prove if something converges, just b/c absval diverges doesn't mean that actual sum diverges

**Example:** Determine whether

$$\sum_{n=1}^{\infty} (-1)^n \quad n^{-1/2} \text{ and } \sum_{n=1}^{\infty} (-1)^n \quad n^{-2}$$

converge or diverge.

Let $a_n = (-1)^n n^{-2}$, then $|a_n| = n^{-2} = \frac{1}{n^2}$ ] p series where $p=2$. $2>1$ so this converges

$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} n^{-2}$ converges as a p-series, so by abs. con. theorem, $\sum_{n=1}^{\infty} a_n$ converges

Let $b_n = (-1)^n n^{-1/2} \to |b_n| = n^{-1/2}$. $\sum_{n=1}^{\infty} |b_n| = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$ diverges (p-series, $p = \frac{1}{2} < 1$). So need to use alternating series test

Let $c_n = n^{-1/2}$ so $\lim_{n\to\infty} n^{-1/2} = 0$ (2nd requirement met). $c_{n+1} = \frac{1}{\sqrt{n+1}}$ ] $c_n = \frac{1}{\sqrt{n}}$ $\overbrace{c_{n+1} \leq c_n}^{c_{n+1} \text{ has bigger denom than } c_n}$

(1st requirement met). So by Alt. series test $\sum_{n=1}^{\infty} (-1)^n b_n$ converges

**Example:** Determine whether

$$\sum_{k=3}^{\infty} (-1)^k \frac{k}{k^2+1} \quad \text{or} \quad \sum_{k=3}^{\infty} (-1)^k \frac{1}{k^2+1}$$

converge or diverge.

$$\sum_{n=1}^{\infty} (-1)^n \frac{1}{n} \Big\} \text{ alternating harmonic series}$$

Let $a_n = \frac{1}{n}$

absval not helpful here as that is a harmonic series which diverges so absval tells us nothing

$$\lim_{n\to\infty} \frac{1}{n} = 0 \Big\} \text{ (2) cond. of Alt Series test met}$$

$$a_{n+1} = \frac{1}{n+1} \leq \frac{1}{n} = a_n \text{ (1) met}$$

So by Alt Series test, $\sum_{n=1}^{\infty} (-1)^n a_n$ converges

L can be 0. Can ignore L notation

**Ratio Test:** → Factorials need ratio test from what we've learned so far

Fewest conditions before this?

- If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L < 1$, then $\sum a_n$ converges.
- If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L > 1$, then $\sum a_n$ diverges.
- If $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = L = 1$, then this test is inconclusive (need to use a different test).

**Example:** Determine whether

$$\sum_{n=1}^{\infty} e^{-n} n!$$

converges or diverges.

why is this $n+1$? [$e^{-n}$, and replace $n$ w/ $n+1$. so $e^{-(n+1)} = e^{-n-1}$

Let $a_n = e^{-n} n!$ Let $a_{n+1} = e^{-n-1}(n+1)!$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{e^{-n-1}(n+1)!}{e^{-n}\, n!}\right| = \frac{e^{-1}\cancel{e^{-n}}(n+1)\cancel{n!}}{\cancel{e^{-n}}\,\cancel{n!}} = |e^{-1}(n+1)|$$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}|e^{-1}(n+1)| = \infty$$

So $a_n$ diverges

**Example:** Determine whether

$$\sum_{n=1}^{\infty} \frac{n^2}{2^n}$$

converges or diverges.

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(n+1)^2}{2^{n+1}} \cdot \frac{2^n}{n^2}\right| = \left|\frac{(n^2+2n+1)\cancel{2^n}}{\cancel{2^n}\,2^1\, n^2}\right|$$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty}\left|\frac{n^2+2n+1}{2n^2}\right| = \frac{1}{2} < 1, \text{ conv.}$$

by ratio test

**Root Test:** ] when entire thing is raised to power of n

- If $\lim_{n\to\infty} |a_n|^{1/n} = L < 1$, then $\sum a_n$ converges.
- If $\lim_{n\to\infty} |a_n|^{1/n} = L > 1$, then $\sum a_n$ diverges.
- If $\lim_{n\to\infty} |a_n|^{1/n} = L = 1$, then this test is inconclusive (need to use a different test).

**Example:** Determine whether

$$\sum_{n=1}^{\infty} \left(\frac{3n+7}{7n+3}\right)^n$$

converges or diverges.

$$|a_n|^{\frac{1}{n}} = \left(\frac{3n+7}{7n+3}\right)^{\frac{n}{n}} = \frac{3n+7}{7n+3} \to \lim_{n\to\infty} \frac{3n+7}{7n+3} = \frac{3}{7} < 1,$$

conv. using root test

**Example:** Determine whether

$$\sum_{k=1}^{\infty} \left(\frac{\ln(n)}{n^2}\right)^n$$

converges or diverges.

$$|a_n|^{1/n} = \left|\frac{\ln(n)}{n^2}\right|^{\frac{n}{n}} \to \lim_{n\to\infty} |a_n|^{\frac{1}{n}} = \lim_{n\to\infty} \left|\frac{\ln(n)}{n^2}\right|$$

By l'hopital, $\lim_{n\to\infty} \frac{1/n}{2n} = \lim_{n\to\infty} \frac{1}{2n^2} = 0$

$0 < 1$, converges by root test